

The Bridge to Post-Quantum: Benchmarking Hybrid Cryptographic Implementation

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Abstract—The advent of fault-tolerant quantum computing threatens the integer-factorization hardness assumptions underpinning RSA and ellipticcurve cryptography. This paper presents a two-part empirical study. In Part 1, we construct gate-level Shor’s algorithm circuits using Qiskit for $N \in 15, 21, 35$ and extrapolate qubit and gate-count requirements to RSA-2048, projecting approximately 4,099–6,144 logical qubits and a wall-clock attack time of under five hours on superconducting hardware ($\tau = 1 \mu s$). In Part 2, we benchmark RSA-2048 KEM, MLKEM-512 (FIPS 203), and a KDF-based Hybrid KEM across 1,000 runs on consumer hardware. ML-KEM-512 achieves key-generation speedups of approximately $5,700\times$ over RSA-2048, and Hybrid KEM encapsulation adds only 0.07 ms of overhead versus standalone MLKEM. These results jointly quantify the quantum threat and confirm that a hybrid KEM bridge is both secure and operationally practical for near-term deployment.

Index Terms—post-quantum cryptography, ML-KEM, RSA-2048, hybrid KEM, Shor’s algorithm, Qiskit, FIPS 203, cryptology

I. INTRODUCTION

The security of virtually all public-key cryptographic infrastructure — from TLS certificates and digital signatures to key-encapsulation mechanisms — rests on the computational hardness of integer factorization (RSA) and the discrete logarithm problem (ECC). Peter Shor’s 1994 polynomial-time quantum algorithm dissolves both assumptions, reducing the factoring of an n -bit modulus to $O(n^3)$ quantum gate operations [2].

No fault-tolerant quantum computer exists today that could break RSA-2048 — but the gap is closing faster than most migration timelines assume. Google’s Willow chip achieving below-threshold error correction (December 2024), IonQ’s 2028 roadmap, and Quantinuum’s fault-tolerant trapped-ion demonstrations — indicate that the transition window is measured in years, not decades. The “harvest now, decrypt later” (HNDL) threat model compounds urgency: adversaries may already be archiving TLSencrypted traffic for future decryption.

In August 2024, NIST released three post-quantum standards: FIPS 203 (ML-KEM, derived from CRYSTALS-Kyber), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) [3]. However, a direct cutover from RSA to PQC is operationally infeasible

for most organizations, which face 12–15+ year migration timelines. Hybrid KEMs — which combine a classical and a post-quantum primitive so that security holds if either component remains unbroken — provide the pragmatic bridge.

Despite strong standardization guidance, the published literature lacks (a) open, gate-level Shor threat models with extrapolations to RSA-2048, and (b) reproducible, statistically robust benchmark data on consumer hardware. This paper closes both gaps with a two-part study: a Qiskit Shor circuit analysis (Part 1) and a 1,000-run empirical benchmark of RSAKEM, ML-KEM-512, and Hybrid KEM (Part 2).

II. BACKGROUND AND RELATED WORK

A. Shor’s Algorithm the and RSA Threat

The security of RSA rests on a single mathematical assumption: that factoring a large composite integer $N = p \times q$ into its prime components is computationally intractable for classical machines. For RSA-2048, the best known classical algorithm — the General Number Field Sieve — requires sub-exponential time, placing a successful attack beyond any realistic classical adversary.

Shor’s algorithm dissolves this assumption. By reducing integer factorization to quantum period-finding via the Quantum Fourier Transform, Shor solves the problem in $O(n^3)$ gate operations — polynomial in the bit-length of N [2]. The implication is not merely theoretical: Gidney and Ekerå (2021) demonstrated that RSA-2048 could be factored using approximately 20 million noisy physical qubits in under eight hours, providing the first concrete resource estimate that moved the threat from algorithmic abstraction to engineering specification [2].

Hardware progress has since compressed that estimate’s assumptions. Google’s Willow processor (December 2024) achieved below-threshold error correction — the foundational milestone required for fault-tolerant quantum computation. IonQ and Quantinuum have published roadmaps projecting fault-tolerant capability within the current decade. Critically, the “harvest now, decrypt later” (HNDL) threat model means adversaries need not wait for full cryptographic breaks: encrypted traffic intercepted today can be archived and decrypted

retroactively once capable hardware exists. The threat is therefore already active, even if not yet fully executable.

B. NIST PQC Standardization

NIST initiated a global post-quantum standardization competition in 2016. After eight years and four evaluation rounds, three standards were finalized in August 2024: FIPS 203 (ML-KEM, derived from CRYSTALS-Kyber) for key encapsulation, FIPS 204 (ML-DSA) for digital signatures, and FIPS 205 (SLH-DSA) as a hash-based signature alternative [3].

ML-KEM is built on the Module Learning With Errors (MLWE) problem, whose hardness is not known to be vulnerable to Shor’s algorithm or any other known quantum attack. It offers three parameter sets — ML-KEM-512, -768, and -1024 — targeting security levels equivalent to AES-128, AES-192, and AES-256 respectively. This study benchmarks ML-KEM-512 specifically because its security level is the closest analogue to the protection RSA-2048 was designed to provide, making it the fair and meaningful comparison point for migration analysis.

Standardization, however, is not the same as deployment. NIST IR 8547 (2024) formalizes the deprecation schedule: RSA-2048 and ECC P-256 are designated for deprecation by 2030, with full disallowance by 2035 [5]. These deadlines establish urgency but do not resolve the operational gap between a policy mandate and a functioning global transition.

C. Hybrid KEMs and Crypto-Agility

The operational gap is significant. Enterprise migration research estimates that large organizations require 12–15 years for complete PQC adoption — encompassing cryptographic inventory, protocol updates, hardware replacement, and certificate infrastructure migration [1]. For medium enterprises the estimate is 8–12 years; for small organizations, 5–7 years. If fault-tolerant quantum computers arrive within the decade, organizations beginning migration today face a window of several years during which neither their classical nor their post-quantum infrastructure is fully secure.

This gap is precisely what hybrid KEMs are designed to bridge. A hybrid KEM runs a classical primitive — typically RSA or ECDH — in parallel with a post-quantum primitive, deriving the shared secret from both via a key derivation function: $ss = \text{KDF}(ss_{\text{classical}} \parallel ss_{\text{PQC}})$. The dual-security property guarantees that the scheme remains secure as long as either component is unbroken — providing quantum-safe protection immediately while preserving classical compatibility during the transition window. Real-world deployment has already begun: Google Chrome adopted X25519+Kyber hybrid key exchange for most TLS connections in 2024, and major certificate authorities began issuing hybrid X.509 certificates in 2025, validating the approach in production environments.

NIST CSWP 39 (2025) formalizes the broader principle as crypto-agility — the capacity to replace cryptographic primitives without disrupting running systems [6]. Crypto-agility and hybrid KEMs are complementary: agility enables the

infrastructure to accommodate the transition; hybrids secure it during the transition.

III. METHODOLOGY

A. Shor’s Circuit Construction

To quantify the quantum threat to RSA concretely, we construct gate-level Shor’s algorithm circuits in Qiskit 2.4.1 for three composite integers $N \in \{15, 21, 35\}$, corresponding to bit-lengths $n = 4, 5$, and 6 . These small- N simulations are deliberately chosen to remain tractable under statevector simulation — which assumes ideal, noiseless qubits — while providing sufficient data points to fit a scaling model for extrapolation to RSA-2048. Each circuit comprises three stages.

1. **QFT/IQFT Construction.** The Quantum Fourier Transform is implemented using Hadamard gates combined with controlled-phase rotations ($\theta = 2\pi/2^k$ for rotation depth k). Bit-reversal swap gates enforce standard qubit ordering. The inverse QFT (IQFT) applies conjugate-negative rotations in reverse sequence. The counting register is allocated $t = 2m$ qubits, where $m = \lceil \log_a N \rceil$, providing sufficient phase resolution for period estimation.
2. **Modular Exponential Oracle.** The oracle U_a encodes modular exponentiation via controlled permutation unitaries $U_a^{2^i}$ acting on the m -qubit work register. For each simulation, the base a is selected as a random integer coprime to N . Trials in which the oracle produces a trivial factor ($\gcd = 1$ or N) are discarded and repeated with a fresh base, consistent with the standard probabilistic structure of Shor’s algorithm.
3. **Measurement and Classical Post-Processing.** Following measurement of the counting register, the observed bit-string is interpreted as a rational approximation to s/r , where r is the unknown period. The denominator of the convergent produced by continued-fraction expansion yields the period estimate r . Prime factors are then recovered via:

$$g_1 = \gcd(a^{r/2} \pm 1, N) \quad (1)$$

Cases where r is odd or where $a^{r/2} \equiv 1 \pmod{N}$ are flagged and measured again. Gate counts and logical qubit requirements are recorded for each simulation. Fitting across $n \in \{4, 5, 6\}$ reveals gate scaling consistent with $2n^3$ and qubit scaling consistent with $3n$, which are applied to extrapolate resource requirements for $N = 2^{2048}$.

B. PQC Benchmarking

All benchmarks were executed on an Intel Core i5- 10th Generation processor, 12 GB RAM, running on Windows, with Python and liboqs. Measurements were conducted under isolated single-threaded conditions to eliminate non-deterministic scheduling overhead from concurrent processes, ensuring that observed latencies reflect cryptographic computation alone.

Three KEM schemes were evaluated. First, RSA-2048 KEM using PKCS1 OAEP, in which encapsulation is defined as RSA public-key encryption of a freshly generated 32-byte random secret, and decapsulation as the corresponding private-key

decryption — providing a direct timing boundary consistent with KEM semantics. Second, ML-KEM-512 from the Open Quantum Safe liboqs library, representing the NIST FIPS 203 standard at the AES-128 equivalent security level. Third, a KDF-based Hybrid KEM in which RSA and ML-KEM operate in parallel: public keys and ciphertexts are concatenated, and the shared secret is derived as

$$ss_{\text{hybrid}} = \text{SHA} - 256(ss_{\text{RSA}} \parallel ss_{\text{ML-KEM}}) \quad (2)$$

This construction provides dual-layer security — the hybrid scheme remains secure as long as either component primitive is unbroken — making it robust against both classical and quantum adversaries simultaneously during the transition window.

Each of the three operations — KeyGen, Encapsulation, and Decapsulation — was measured over 1,000 independent runs. Prior to measurement, 20 warmup iterations were discarded to bring CPU cache and runtime environments to steady state. Timing used high-resolution wall-clock measurements at microsecond precision via Python’s `time.perf_counter()`. Results are reported as mean across the 1,000 runs.

IV. RESULTS

A. Shor Circuit Projections

Table I summarizes simulated and projected qubit and gate requirements. For $N = 15$ ($n = 4$), the circuit requires 12 logical qubits and approximately 128 gates. Fitting a $2n^3$ model across the three data points yields the RSA-2048 projection

TABLE I
SHOR CIRCUIT PROJECTIONS

N	Bits (n)	Qubits	Gates (Approx.)	Model
15	4	12	128	Sim.
21	5	15	250	Sim.
35	6	18	432	Sim.
2^{2048}	2048	4,099 - 6144	1.72×10^{10}	Extrap.

The empirical (naïve) projection yields 6,144 logical qubits; the optimized theoretical model yields 4,099, consistent with prior literature estimates of approximately 4,000 logical qubits. Runtime projections for RSA-2048: ≈ 4.77 hours on superconducting hardware ($\tau = 1 \mu\text{s}$ gate time) and ≈ 19.88 days on trapped-ion hardware ($\tau = 100 \mu\text{s}$). Both figures assume logical (error-corrected) gate times; physical qubit overhead from surface codes would increase these estimates substantially.

B. PQC Benchmark Results

Table II presents 1,000-run benchmark results across the three KEM schemes.

ML-KEM-512 is approximately 5,700× faster than RSA-2048 at key generation, 2.4× faster at encapsulation, and 30× faster at decapsulation. The hybrid scheme’s encapsulation latency of 0.2443 ms represents the additive cost of running both primitives in parallel — approximately 0.07 ms over standalone ML-KEM — confirming negligible overhead.

TABLE II
1000 RUN KEM BENCHMARK (MEAN $\pm \sigma$, MS)

Operations	RSA-2048	MEL-KEM-512	Hybrid KEM
KeyGen (ms)	1150 ± 12.30	0.0151 ± 0.0034	-
Encap (ms)	0.1705 ± 0.0133	0.0716 ± 0.0076	0.2443 ± 0.0189
Decap (ms)	1.9961 ± 0.0811	0.0673 ± 0.0069	~ 2.06

C. Transmission Size Analysis

Table III compares transmission sizes relevant to protocol integration.

TABLE III
TRANSMISSION SIZE COMPARISON (BYTES)

Scheme	Pub. Key (B)	Ciphertext (B)	Total (B)
RSA-2048	294	256	550
ML-KEM-512	800	768	1568
Hybrid KEM	1094	1024	2118

ML-KEM-512’s larger key and ciphertext sizes (800 B and 768 B respectively) reflect its lattice-based structure. The Hybrid KEM’s total transmission footprint of approximately 2.1 KB represents a manageable increase for TLS handshakes and certificate exchanges — well within the constraints of modern network MTUs.

V. DISCUSSION

A. Threat Quantification and Implication

Our Shor projections confirm that breaking RSA-2048 requires approximately 4,099 logical qubits — a target reachable within current decade hardware roadmaps. The 4.77-hour superconducting runtime assumes error-free logical qubits; accounting for surface-code overhead (approximately 1,000 physical qubits per logical qubit at current error rates) raises the physical qubit requirement toward the millions, consistent with Gidney and Ekerå [2]. However, rapid progress in error correction — exemplified by Google Willow’s below-threshold demonstration — compresses this timeline in ways that 2021 estimates did not anticipate.

The HNDL threat, however, makes the question of when FTQCs arrive less critical than when migration is complete. Organizations archiving TLS traffic today need RSA-2048 deprecated before decryption becomes feasible — not after.

B. Hybrid KEM as a Practical Bridge

Our benchmarks show that the hybrid KEM construction imposes latency overhead of approximately 0.07 ms at encapsulation relative to standalone ML-KEM-512. This is operationally indistinguishable from RSA-only latency in any network context and is negligible relative to typical TLS handshake round-trip times (≥ 10 ms). The dual-security

property — that the hybrid scheme is broken only if both RSA and ML-KEM are simultaneously broken — provides a stronger security guarantee than either alone during the transition window.

The 2.1 KB total transmission footprint of the hybrid scheme warrants attention in highly constrained environments (IoT, satellite links), but is unproblematic for enterprise TLS, VPN, and API contexts, which this study targets.

C. Limitations

This study benchmarks on a single hardware configuration; multi-core, ARM, and constrained device results may differ. The Shor simulation uses ideal (noiseless) Qiskit statevector simulation; real device execution with error mitigation is beyond this study's scope. Physical qubit overhead from error correction is not modeled.

VI. CONCLUSION & FUTURE WORK

This paper provides a joint empirical answer to two open questions in the PQC migration literature: how quantifiable is the Shor threat to RSA-2048, and how practical is a hybrid KEM bridge?

On the threat side, our gate-level Shor models project RSA-2048 vulnerability at approximately 4,099 – 6,144 logical qubits and under five hours on superconducting hardware — a near-term risk under current hardware trajectories. On the migration side, ML-KEM-512 outperforms RSA-2048 by 5,700× in key generation and 30× in decapsulation, and a KDF based Hybrid KEM adds only 0.07 ms of encapsulation overhead, confirming immediate deployability.

Together, these findings support a two-phase transition strategy: (1) deploy Hybrid KEM immediately to achieve HNLD protection and dual security guarantees while maintaining backward compatibility, and (2) deprecate the RSA component on a planned schedule aligned with NIST IR 8547's 2030 deadline.

Future work will model physical error-correction overhead using surface codes, test PQC performance under constrained network topologies and IoT hardware, evaluate ML-KEM-768 and ML-KEM-1024 for higher-security environments, investigate multithreaded Hybrid KEM encapsulation to reduce sequential overhead, and extend the benchmark methodology to aggregate results across multiple independent sessions — reporting minimum, maximum, and mean-of-means to capture session-to-session variability that single-run evaluations cannot expose.

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